

Products, liquidity, and price stability in the EEX power futures market:

Evidence from German, French, and Italian power futures

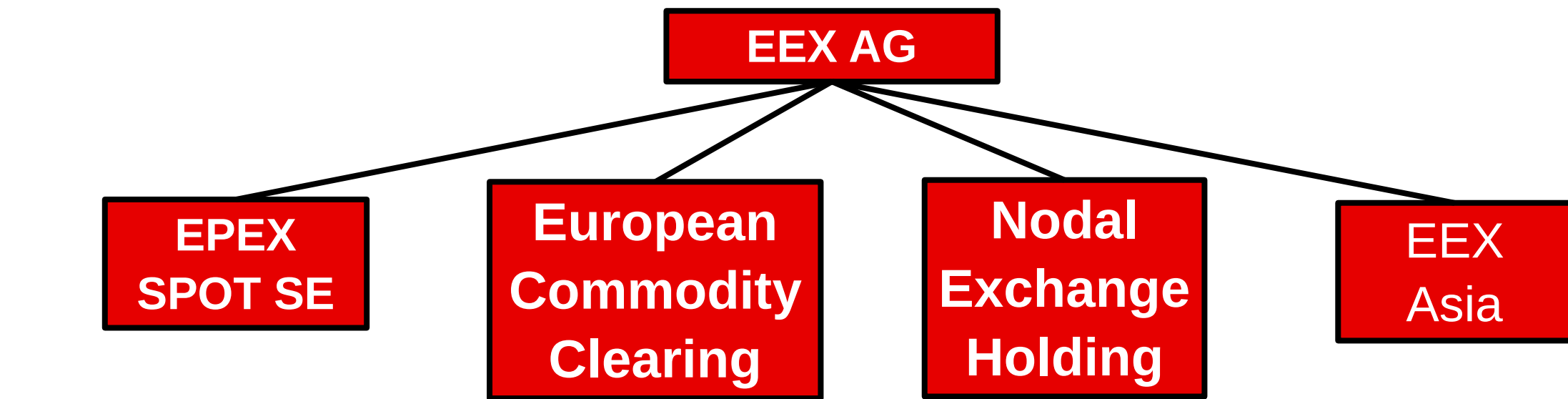
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Introduction

The European Energy Exchange (EEX) is a leading energy exchange which builds secure, successful and sustainable commodity markets worldwide.

The EEX corporate pillar

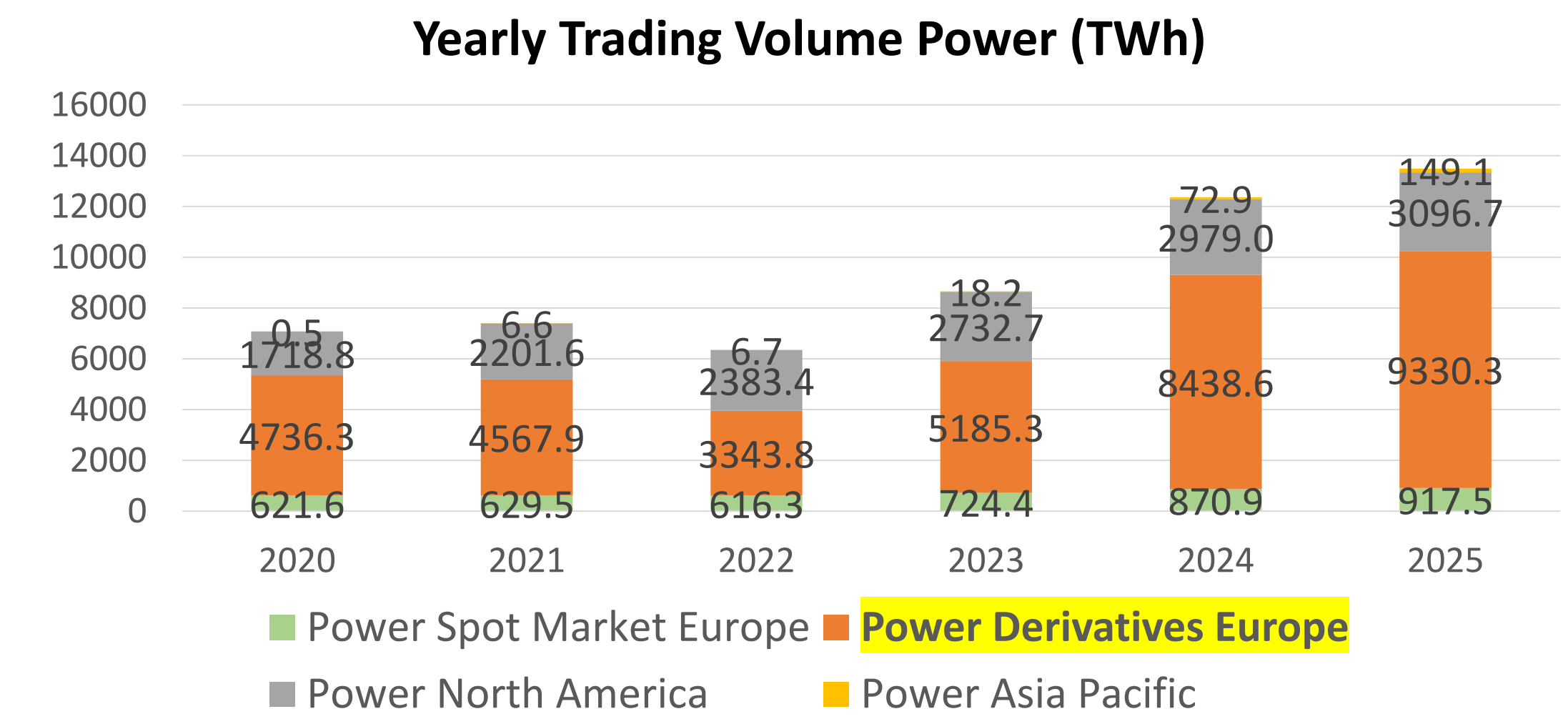


The EEX market

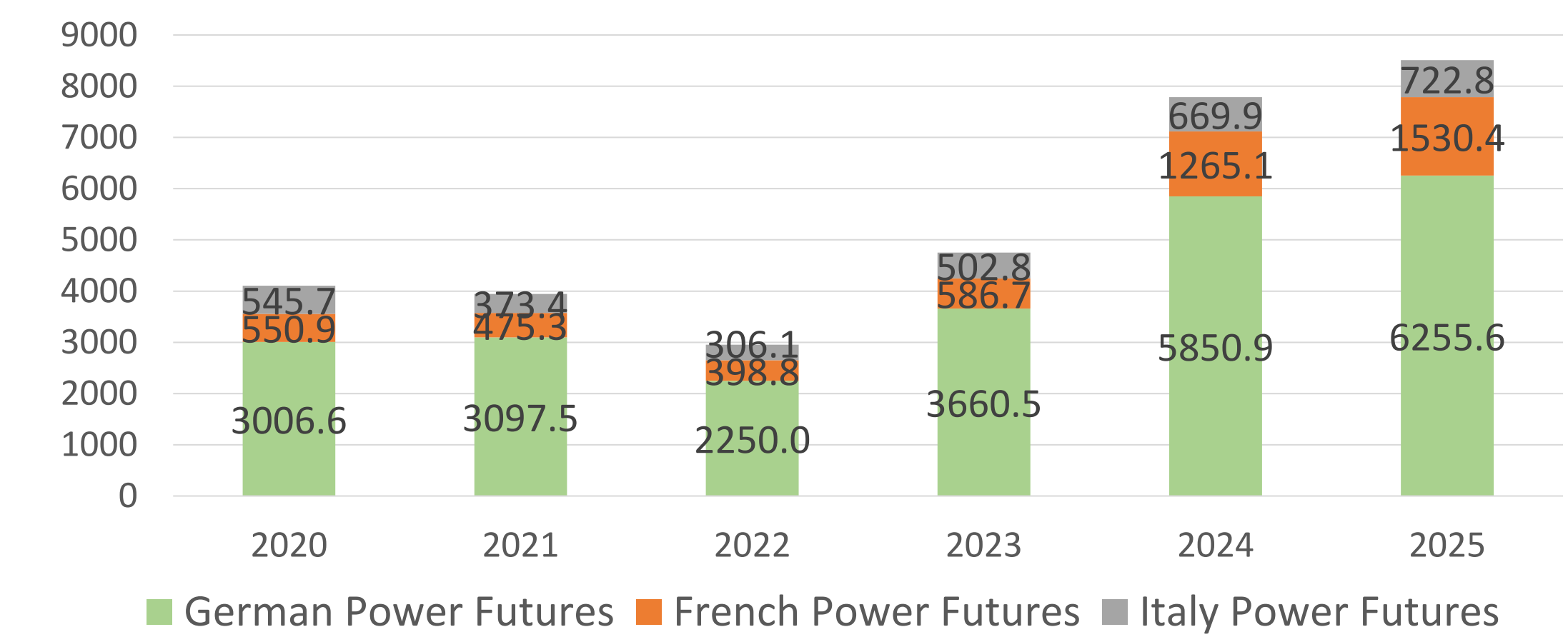


Trading participants fall into five major categories:

- Energy Traders:** RWE Supply & Trading, Uniper Global Commodities SE, Vattenfall Energy Trading GmbH, E.ON Energy Market GmbH
- Energy Producers:** Statkraft Market GmbH and BayWa r.e. Energy GmbH
- Commercial Utilities/ Suppliers:** All Stadtwerke (Stadtwerke Muenchen, Leipzig, Duesseldorf, Giessen, Flensburg), Enercity AG
- Financial Institution:** Citigroup Global Markets Europe AG, Commerzbank AG, J.P Morgan SE, and Morgan Stanley Europe
- Grid Operators:** 50Hertz Transmission GmbH, Transnet BW GmbH
- Transport and Logistics:** Deutsche Lufthansa Aktiengesellschaft, DB Energy GmbH



The trading volume of the most liquid power futures contract in Europe (TWh)



Source of the figure data: EEX Group Annual Volumes Reports

Objectives

- To understand the European Energy Exchange market structure, product, and market participants
- To investigate the volatility dynamic of the **most liquid European power futures**, namely German power future, France power futures, and Italian power futures
- To investigate the period of economic distress impacting the future price stability on the most liquid European power futures

Result

Table 1. Descriptive statistics of futures returns

Statistics	German Power	French Power	Italian Power
Mean	0.00015	-0.00004	0.00030
Variance	0.00075	0.00152	0.00110
Minimum	-0.187	-0.429	-0.302
Maximum	0.275	0.423	0.333
Skewness	1.163	0.670	0.290
Kurtosis	19.772	27.079	16.227
Jarque-Bera	52317*	74517*	22993*
ADF	-14.398*	-13.030*	-13.837*
KPSS	0.068	0.066	0.064
LB-Q [25]	58.099*	100.225*	51.799*
ARCH LM test [25]	133.73*	59.617*	512.252*

Note: * indicates the statistical significance at the 1% significance level

Table 2. Parameter Estimation of GARCH (1,1) model

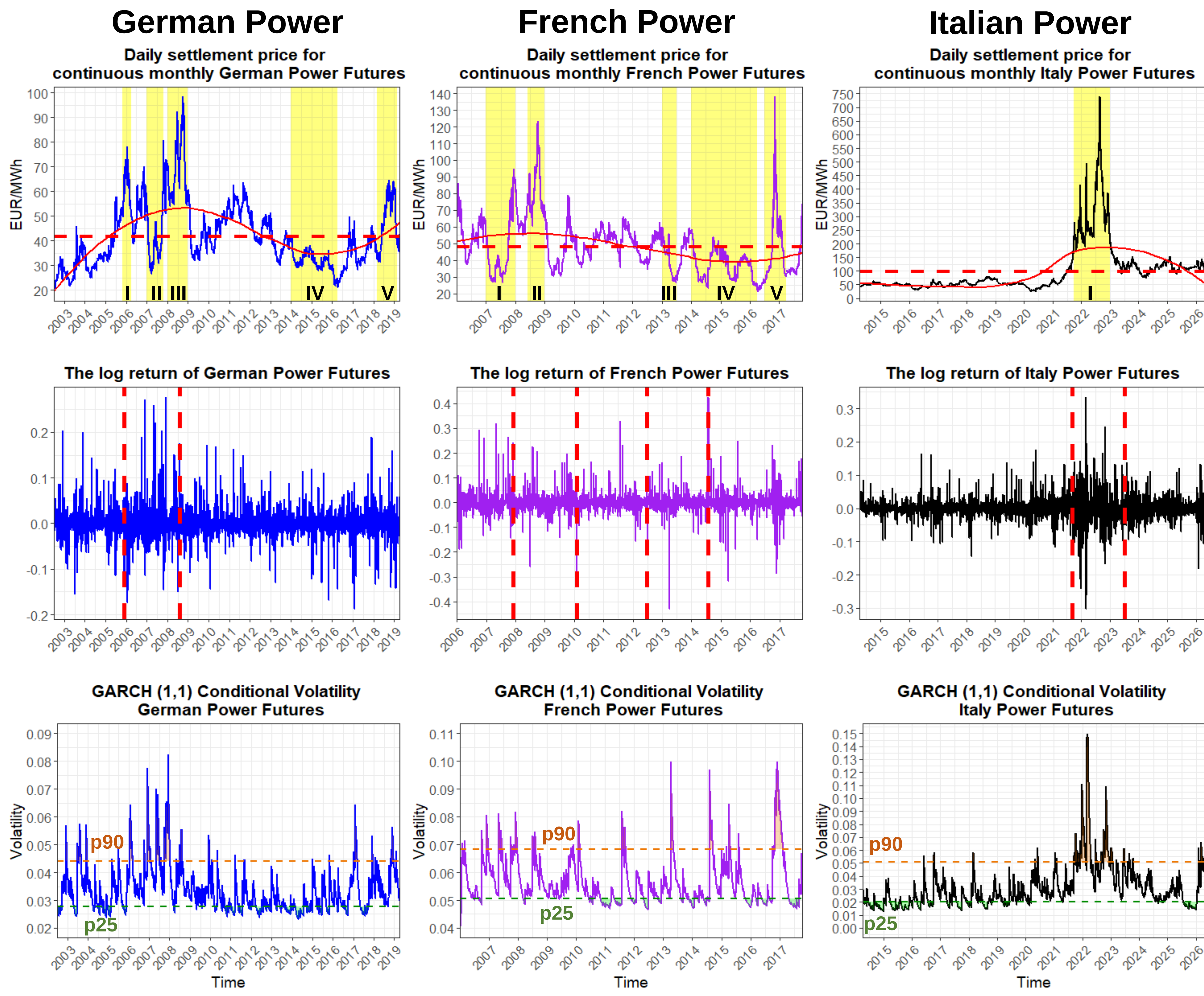
Parameters	German Power	French Power	Italian Power
μ	-0.0060**	-0.00083**	-0.00023 ns
Ω	0.00001*	0.00008*	0.00001*
α (ARCH)	0.530*	0.038*	0.09*
β (GARCH)	0.945*	0.956*	0.90*
Shape	2.345*	2.170*	3.19*

Note: * and ** indicates the statistical significance at 1% and 5 % significance level, respectively. ns represents "not significant"

Table 3. Diagnostic test of the GARCH (1,1) model

Parameters	German Power	French Power	Italian Power
LB-Q [10]	9.47 ns	17.42 ns	8.56 ns
LB-Q*2 [10]	9.26 ns	2.94 ns	7.57 ns

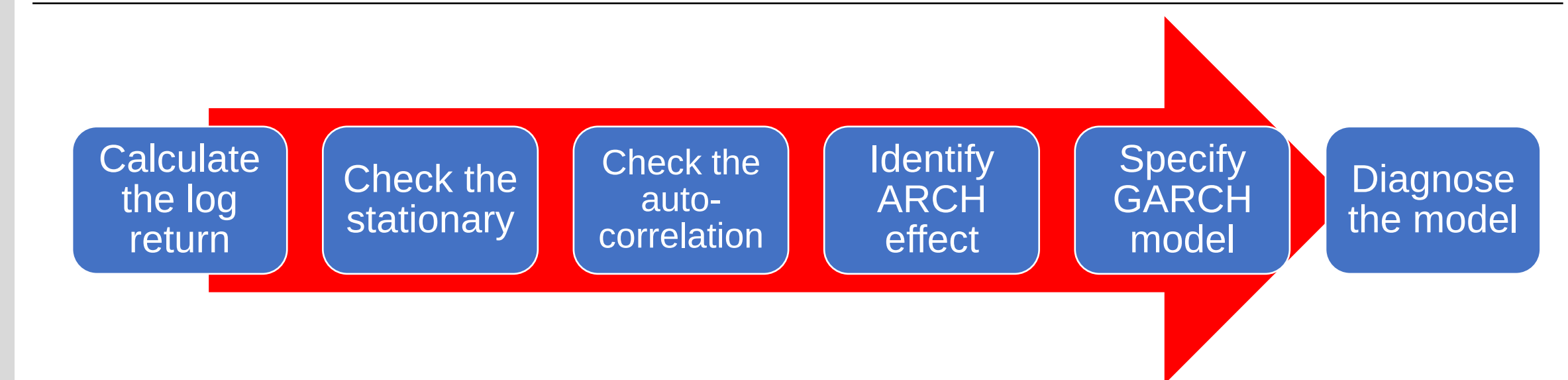
Note: ns represents "not significant"



Data and Methodology

Source of data: Refinitive datastream

Power Futures	Ticker in Refinitive	Periods
German Power Futures	LFUTEBMC	2002 - 2019 (18 years)
French Power Futures	EFBCS00	2006 - 2017 (12 years)
Italian Power Futures	LFUTEXMC	2014 - 2026 (13 years)



Discussion and Conclusion

German Power Future

No.	Period	Drivers
I	Early 2006	Cold winter: colder than average in part of Europe Increase in Coal price: German electricity power rely on coal Geopolitical: Russian gas export temporarily disturbed
II	2007	Carbon market crash: European government issued pollution permit based on historic industry rather than verified emission data.
III	2008	Financial crisis and Greek government debt crisis
IV	2014 - 2015	Geopolitical: Russia annex Crimea The global oil and commodity crash: OPEC decided not to cut oil production despite a global economic slowdown. This pull down the entire energy market.
V	2018 - 2019	Soaring carbon price and European summer drought

French Power Future

No.	Period	Drivers
I	2007	Carbon market crash: similar to German Power Future
II	Mid 2008	Financial crisis and Greek government debt crisis
III	Mid 2013	Macroeconomic stagnation: Industrial slowing down The renewable energy boom: Massive subsidized investment in wind and solar generation
IV	2014 - 2015	Similar to what occurred with German Power Futures
V	Mid 2016 - early 2017	Nuclear safety crisis: French Nuclear Safety Authority shut down and extend maintenance up 18 nuclear power plant. 20% of the French nuclear power plant capacity was unavailable

Italian Power Future

No.	Period	Drivers
I	Late 2021 - 2023	Geopolitical: the peak future price was driven by Russia's Invasion of Ukraine The Nord Stream 1 shutdown: the extreme scarcity of physical gas that being used to run Italian generators Severe European drought: Heatwave restricted river flows and reduce the Italy's domestic hydroelectric output and disrupted cooling water loops for thermal plants